

LAB EXPERIMENTS

Q2) Write a C program for monoalphabetic substitution cipher maps a plaintext alphabet to a ciphertext alphabet, so that each letter of the plaintext alphabet maps to a single unique letter of the ciphertext alphabet.

Aim

To write a C program to implement the **Monoalphabetic Substitution Cipher**, where each letter of the plaintext alphabet is replaced with a unique corresponding letter from a predefined ciphertext alphabet.

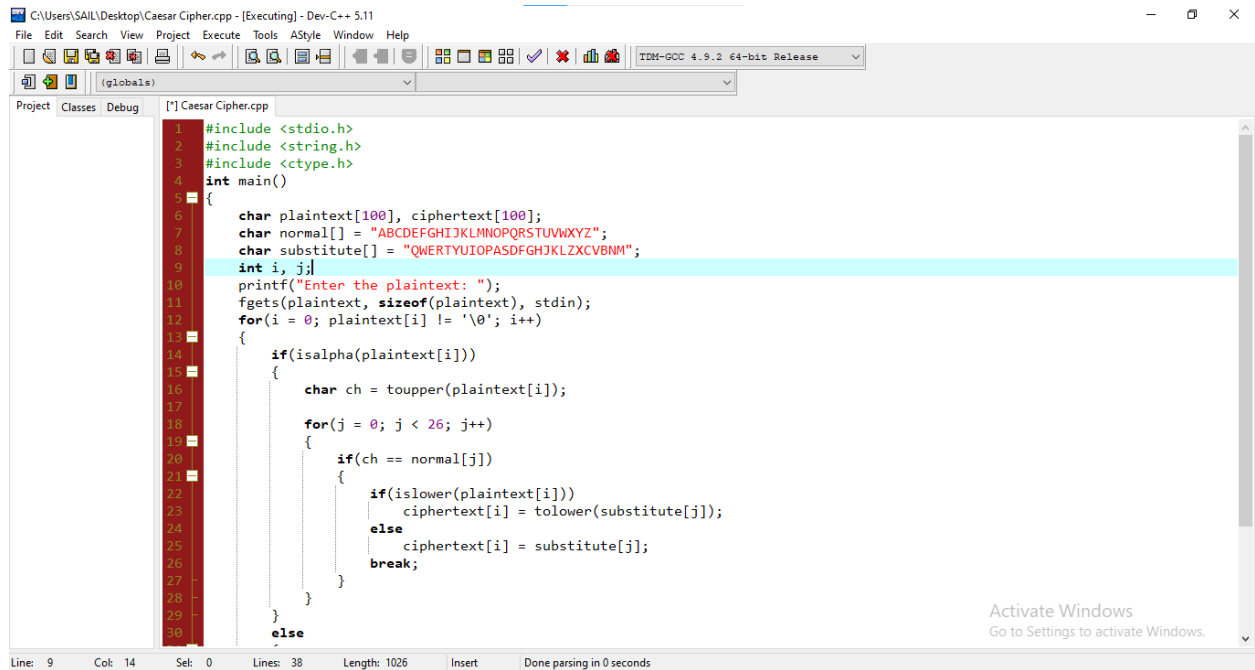
Algorithm

1. Start the program.
2. Declare arrays for the plaintext, ciphertext, normal alphabet, and substitution alphabet.
3. Initialize the normal alphabet as "ABCDEFGHIJKLMNOPQRSTUVWXYZ".
4. Initialize a substitution alphabet with a unique permutation of letters (e.g., "QWERTYUIOPASDFGHJKLZXCVBNM").
5. Read the plaintext message from the user.
6. Traverse each character of the plaintext.
7. If the character is an uppercase or lowercase alphabet, find its corresponding position in the normal alphabet and replace it with the mapped character from the substitution alphabet. Leave non-alphabetic characters unchanged.
8. Store the substituted characters in the ciphertext.
9. Display the encrypted ciphertext.
10. Stop the program.

Result

Thus, the C program to implement the **Monoalphabetic Substitution Cipher** was successfully executed, and the corresponding encrypted ciphertext was obtained.

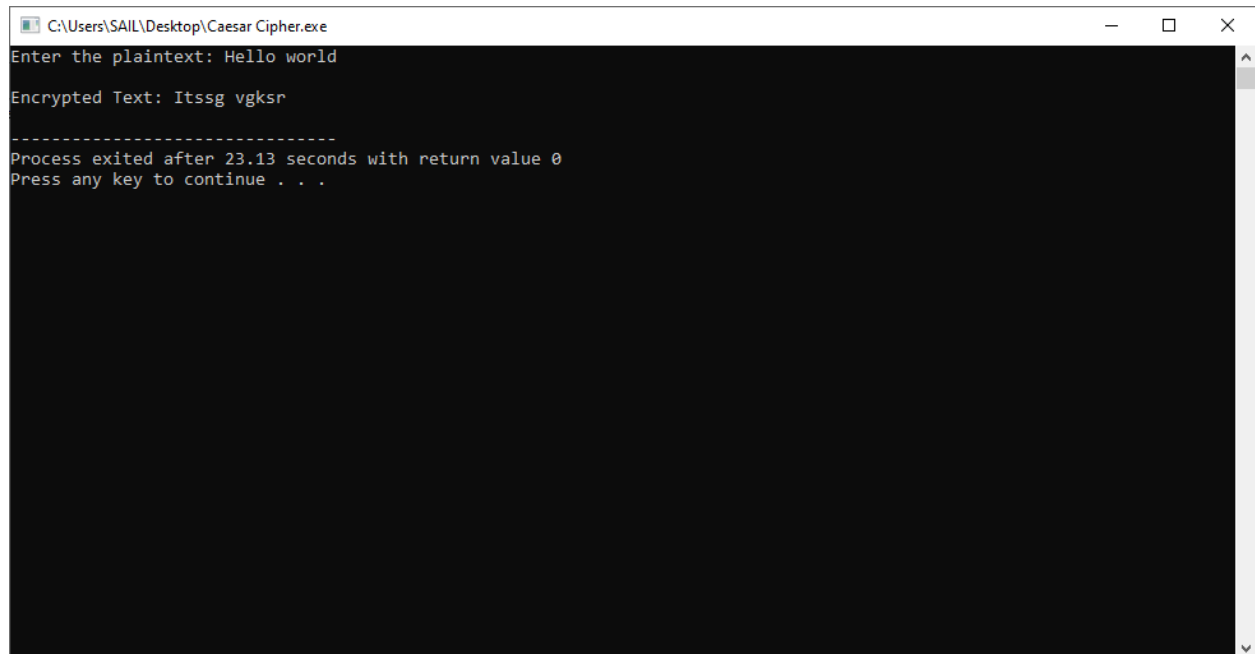
Code:



The screenshot shows a C++ IDE with the file "Caesar Cipher.cpp" open. The code implements a Caesar cipher encryption. It includes `<stdio.h>`, `<string.h>`, and `<ctype.h>`. The `main` function prompts the user to enter a plaintext string. It then iterates through each character of the plaintext. If the character is an uppercase letter, it is converted to lowercase and then shifted based on a predefined substitution key. The substitution key is a 26-character string: "QWERTYUIOPASDFGHJKLZXCVBNM". The encrypted text is stored in the `ciphertext` array. The status bar at the bottom indicates the current line is 9, column is 14, and the total length of the file is 1026 characters.

```
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4 int main()
5 {
6     char plaintext[100], ciphertext[100];
7     char normal[] = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";
8     char substitute[] = "QWERTYUIOPASDFGHJKLZXCVBNM";
9     int i, j;
10    printf("Enter the plaintext: ");
11    fgets(plaintext, sizeof(plaintext), stdin);
12    for(i = 0; plaintext[i] != '\0'; i++)
13    {
14        if(isalpha(plaintext[i]))
15        {
16            char ch = toupper(plaintext[i]);
17
18            for(j = 0; j < 26; j++)
19            {
20                if(ch == normal[j])
21                {
22                    if(islower(plaintext[i]))
23                        ciphertext[i] = tolower(substitute[j]);
24                    else
25                        ciphertext[i] = substitute[j];
26                    break;
27                }
28            }
29        }
30    }
31    else
32    {
33        ciphertext[i] = plaintext[i];
34    }
35 }
```

Output



The screenshot shows the output of the Caesar Cipher program. The user entered "Hello world" as the plaintext. The program output shows the encrypted text as "Itssg vgksr". The program then exits with a return value of 0 and prompts the user to press any key to continue.

```
C:\Users\SAIL\Desktop\Caesar Cipher.exe
Enter the plaintext: Hello world

Encrypted Text: Itssg vgksr

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Process exited after 23.13 seconds with return value 0
Press any key to continue . . .
```